SAMA

Solar Alone Multi-Objective Advisor V2

Comprehensive User Documentation | v1

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0. Before You Start: Installing Python

SAMA requires Python 3.9 or higher. If you do not already have Python installed on your computer, follow the steps below before proceeding to any other section of this guide.

0.1 Checking if Python is Already Installed

Open a terminal (Command Prompt on Windows, Terminal on macOS/Linux) and type:

```
python --version
```

If you see a version number such as Python 3.11.4, Python is already installed and you can skip this section. If you see an error or a version below 3.9, follow the steps below.

0.2 Installing Python on Windows

Go to https://www.python.org/downloads/ and click Download Python (choose the latest 3.x version).
Run the installer. On the first screen, check the box that says "Add Python to PATH" before clicking Install Now. This is the most important step - without it, Python commands will not work in the terminal.
Once installation is complete, close and reopen your Command Prompt and verify with python --version

0.3 Installing Python on macOS

Go to https://www.python.org/downloads/ and download the macOS installer.
Run the .pkg installer and follow the on-screen steps.
Open Terminal and verify with: python3 --version

0.4 Opening a Terminal

Throughout this guide, instructions refer to running commands in a terminal. Here is how to open one on each platform:
Windows: Press the Windows key, type cmd, and press Enter. Or open File Explorer, navigate to your project folder, click the address bar, type cmd, and press Enter to open a terminal directly in that folder.
macOS: Press Command + Space, type Terminal, and press Enter.
Linux: Press Ctrl + Alt + T.

0.5 Verifying pip

pip is the Python package installer and is included automatically with Python 3.9 and later. Verify it is available:

```
pip --version
```

If pip is not found, run: python -m ensurepip –upgrade

1. OVERVIEW

SAMA (Solar Alone Multi-Objective Advisor) is an open-source Python platform for the design, sizing, and operation optimization of hybrid renewable energy microgrids. It performs location-specific techno-economic assessment using site-dependent electrical and thermal load profiles, meteorological data, component degradation models, and utility-specific billing structures.

Unlike proprietary tools such as HOMER Pro and HOMER Grid, which are now closed-source and license-restricted, SAMA provides transparent optimization algorithms, fully extensible architecture, and no cost barriers for researchers, practitioners, and resource-constrained institutions.

1.1 What SAMA Optimizes

SAMA finds the optimal combination of five design variables:

- Number of PV modules (Npv) in units of rated PV capacity (kW)
- Number of wind turbines (Nwt)
- Number of battery packs (Nbat)
- Number (size) of diesel generator units (N_DG)
- Inverter capacity (Cn_I) in kW

The search space is bounded by VarMin and VarMax (default: 0 to 60 for each variable). The optimizer explores this space to minimize the primary objective, Net Present Cost (NPC), while satisfying reliability, renewable fraction, budget, and rooftop-area constraints. When EM is set to 1, emissions (Levelized Emissions, LEM) are included as a second objective.

1.2 Key Capabilities

Capability	Detail
System components	Solar PV, wind turbines, battery storage (lead-acid or lithium-ion), diesel generator, utility grid, air-source heat pump (Bosch or Goodman), electric vehicle (V2X)
Optimization algorithms	Particle Swarm Optimization (PSO), Advanced Differential Evolution (ADE), Artificial Bee Colony (ABC), Grey Wolf Optimizer (GWO)
Energy management	8,760-hour hourly dispatch with EMS, EV smart charging/discharging with V2X lookahead arbitrage, heat pump thermal-electric coupling
Electricity tariffs	8 rate structures: flat, seasonal, monthly, tiered, seasonal tiered, monthly tiered, Time-of-Use (TOU), Ultra-Low TOU (ULO)
Natural gas tariffs	8 rate structures for NG: flat, seasonal, monthly, tiered, seasonal tiered, monthly tiered, PG&E G-1 therms-based, Enbridge EGD 4-tier m3-based
Financial analysis	25-year Net Present Cost, LCOE, levelized emissions, escalation rates, RE incentives (e.g. 30% federal tax credit), net metering reconciliation
Load inputs	10 electrical load input modes (CSV, monthly/annual averages, generic profiles), thermal load from Excel

Weather inputs	SAM NSRDB API integration, user CSV files, monthly/annual averages
Outputs	Convergence plots, cash flow charts, energy distribution, battery SOC, grid interconnection, EV energy charts, HP performance charts, hourly data CSV
License	GNU General Public License v3.0 (GPL-3.0), free to use, modify, and distribute

1.3 Comparison to HOMER Pro

SAMA has been validated against HOMER Pro in diverse climatic zones (Sacramento, California and New Bern, North Carolina) and achieves close agreement in optimal system configurations and performance metrics. SAMA additionally supports constraint scenarios such as rooftop area limits, net-metering caps, renewable fraction requirements, and loss-of-power-supply probability, which are handled more flexibly than in proprietary tools.

2. THREE WAYS TO USE SAMA

SAMA can be used in three different ways depending on your background and needs:

Method	Who It Is For	How to Get It
Method 1: Raw Python Code	Researchers, developers, and advanced users who want to modify algorithms or parameters directly in the source code	Download from GitHub and open in a Python IDE such as PyCharm
Method 2: Python Package (pip)	Users who want to run SAMA as a command-line tool or use it in their own Python scripts without modifying internals	pip install sama (available as the SAMA Py Package)
Method 3: Windows .exe Application (Alpha)	Non-technical users who want a point-and-click graphical user interface without installing Python. Note: this version is currently in the alpha development phase.	Available on request from the developers

3. METHOD 1 - USING THE RAW PYTHON CODE

This method gives you full access to all source files. You can directly edit Input_Data.py to set your parameters and run the optimizer scripts directly. This is the approach used in the examples/ folder.

3.1 Prerequisites

- Python 3.9 or higher
- A Python IDE such as PyCharm (recommended), VS Code, or Spyder
- Git (optional, for cloning the repository) or a browser to download the ZIP from GitHub

3.2 Downloading and Opening the Code

The SAMA v2.0.1 source code is hosted on GitHub at the following location:

GitHub Repository (Method 1 source code):
https://github.com/Sas1997/SAMA/tree/main/Backend%20Codes/SAMA%20V2.0.1-GitHub

1. Open the URL above in your browser.
2. Click the green Code button and choose Download ZIP, or clone with Git:
   `git clone https://github.com/Sas1997/SAMA.git`
3. Extract the ZIP (or navigate into the cloned folder) and open the folder Backend Codes/SAMA V2.0.1-GitHub in PyCharm (or your preferred IDE).
4. PyCharm will detect the project structure automatically. Set up a Python interpreter (Python 3.9 or higher) in File > Settings > Project > Python Interpreter.
5. Install the required packages. You can do this in the PyCharm terminal or any command prompt:

```
pip install numpy pandas scipy numba matplotlib openpyxl seaborn numpy-financial questionary PyYAML
```

6. Verify the setup by running:

```
python -c "from sama.core.Input_Data import InData; print('OK')"
```

3.3 Configuring Parameters Directly in Input_Data.py

Open sama/sama/core/Input_Data.py in your IDE. All parameters are set in the Input_Data.__init__ method. The key sections to edit are described below.

3.3.1 System Components

Enable or disable each component by setting its flag to 1 (included) or 0 (excluded):

```
self.PV = 1        # Solar PV
self.WT = 0        # Wind turbine
self.DG = 1        # Diesel generator
self.Bat = 1       # Battery storage
self.Grid = 1      # Utility grid connection
self.HP = 1        # Heat pump (Goodman or Bosch)
self.EV = 0        # Electric vehicle
self.Lead_acid = 0  # Lead-acid battery
self.Li_ion = 1     # Li-ion battery
```

3.3.2 Optimization Search Space

VarMin and VarMax define the lower and upper bounds for the five design variables [Npv, Nwt, Nbat, N_DG, Cn_I]:

```
self.VarMin = np.array([0, 0, 0, 0, 0])      # lower bounds
self.VarMax = np.array([60, 60, 60, 20, 60]) # upper bounds
```

MaxIt, nPop, and Run_Time control the optimizer:

```
self.MaxIt = 200    # maximum number of iterations
self.nPop = 50      # population size (number of candidate solutions)
self.Run_Time = 1   # number of independent optimization runs
```

3.3.3 Load Input

Set load_type to choose how the electrical load is provided:

load_type value	Description	Required variable
1	Hourly load from a CSV file (8,760 rows)	path_Eload: path to CSV
2	Monthly hourly average load	Monthly_haverage_load: array of 12 values (kW)
3	Monthly daily average load	Monthly_daverage_load: array of 12 values (kWh/day)

4	Monthly total load	Monthly_total_load: array of 12 values (kWh/month)
5	Generic profile scaled to monthly totals	user_defined_load: array of 12 monthly totals
6	Annual hourly average	Annual_haverage_load: single value (kW)
7	Annual daily average	Annual_daverage_load: single value (kWh/day)
8	Generic profile scaled to annual total	Annual_total_load: single value (kWh/year)
9	Exact generic load profile (unscaled)	peak_month: 'July' or 'January'
10	Generic profile scaled to user daily profile from CSV	path_Eload_daily: path to 24-row CSV

Note: The content/ folder contains a sample Eload.csv for Boston, MA. To use your own load, replace this file or set path_Eload to your file path.

3.3.4 Weather Input

G_type selects irradiance input, T_type selects temperature, and WS_type selects wind speed:

Type flag	Value 1	Value 2	Value 3	Value 4
G_type (irradiance)	From NSRDB via SAM API (uses METEO.csv)	User CSV file (plane-of-array)	N/A	N/A
T_type (temperature)	From NSRDB via SAM API (uses METEO.csv)	User CSV file	Monthly averages (12-value array)	Annual average (single value)
WS_type (wind speed)	From NSRDB via SAM API (uses METEO.csv)	User CSV file	Monthly averages (12-value array)	Annual average (single value)

The default weather_url points to the bundled METEO.csv, which contains sample data for London, Ontario from the National Solar Radiation Database. Key PV geometry parameters:

```
azimuth = 180   # degrees (180 = south-facing)
tilt    = 33    # degrees (tilt angle of PV modules)
soiling = 5     # percent soiling losses
```

3.3.5 Economic Parameters

Parameter	Default	Description
n	25	Project lifetime in years
n_ir_rate	2.75	Nominal discount rate (%)

e_ir_rate	2.0	Expected inflation rate (%)
Budget	2,000,000	Maximum allowable capital cost ($)
Tax_rate	0	Equipment sales tax (%)
RE_incentives_rate	30	Renewable energy incentives / federal tax credit (%)
Pricing_method	2	1 = top-down (total system $/kW), 2 = bottom-up (itemized costs)

3.3.6 Component Cost Parameters (Bottom-Up, Pricing_method = 2)

Component	Capital cost param	Replacement param	O&M param	Units
PV modules	C_PV = 338	R_PV = 338	MO_PV = 30.36	$/kW, $/kW, $/kW/year
Inverter	C_I = 314	R_I = 314	MO_I = 0	$/kW, $/kW, $/kW/year
Wind turbine	C_WT = 1200	R_WT = 1200	MO_WT = 40	$/kW, $/kW, $/kW/year
Diesel generator	C_DG = 818	R_DG = 818	MO_DG = 0.016	$/kW, $/kW, $/op.h
Battery	C_B = 1450	R_B = 1450	MO_B = 10	$/kWh, $/kWh, $/kWh/year
Heat pump	C_HP = 109.5	R_HP = 109.5	MO_HP = 20	$/1000 BTU/hr, $/1000 BTU/hr, $/year
EV battery replacement	N/A	R_EVB = 27000	MO_EV = 0	$, $/year

Additional PV engineering costs (bottom-up) are summed into Engineering_Costs ($/kW):

```
self.Fieldwork = 178          # $/kW
self.Officework = 696         # $/kW
self.Electrical_BoS = 333     # $/kW
self.Structrual_BoS = 237     # $/kW
```

3.3.7 Electricity Tariff Structure

Set rateStructure to choose the utility tariff model for grid electricity purchases:

rateStructure	Type	Required parameters
1	Flat rate	flatPrice: single $/kWh value
2	Seasonal rate	seasonalPrices [summer, winter], season [12-value month array]
3	Monthly rate	monthlyPrices: 12-value array ($/kWh per month)
4	Tiered rate	tieredPrices, tierMax: tier prices and monthly kWh limits

5	Seasonal tiered rate	seasonalTieredPrices [2 seasons x 3 tiers], seasonalTierMax, season
6	Monthly tiered rate	monthlyTieredPrices [12 x 3], monthlyTierLimits [12 x 3]
7	Time-of-Use (TOU)	onPrice, midPrice, offPrice [summer, winter], onHours, midHours, season, treat_special_days_as_offpeak
8	Ultra-Low TOU (ULO)	onPrice, midPrice, offPrice, ultraLowPrice [summer, winter], onHours, midHours, ultraLowHours, season, treat_special_days_as_offpeak

Grid sell price structure is set separately with sellStructure:

sellStructure	Description	Parameter
1	Flat sell price	Csell = np.full(8760, price): one price all year
2	Monthly sell price	monthlysellprices: 12-value array ($/kWh per month)
3	Same as buy price	Csell = Cbuy: NEM at retail rate

Net metering is controlled by NEM (1 = enabled) and NEM_fee (one-time connection fee in $). When NEM = 1, surplus generation credits are reconciled annually.

3.3.8 Natural Gas Tariff (for Heat Pump)

When HP = 1 and a natural gas furnace baseline is being compared, set rateStructure_NG to one of 8 structures analogous to the electricity rate structures. The NG price is entered in the user's native unit ($/m3, $/therm, or $/kWh) and SAMA converts internally using NG_energycontent (10.81 kWh/m3) and Furnace_eff (0.94).

3.3.9 System Constraints

Parameter	Default	Description
LPSP_max_rate	0.000999	Maximum Loss of Power Supply Probability (LPSP) as a percentage. The system must meet at least (100 minus LPSP_max_rate)% of demand.
RE_min_rate	50	Minimum renewable energy fraction required (%). System must have at least this fraction from PV and WT.
EM	0	Objective mode. 0 = minimize NPC only. 1 = minimize NPC + Levelized Emissions (LEM).
Budget	2,000,000	Maximum total capital cost ($). Infeasible solutions above this are penalized.
cap_option	2	Capacity sizing constraint. 1 = explicit cap in kW (cap_size). 2 = size to annual load. 3 = rooftop area limit. 4 = no limit.

3.4 Running the Optimization

After configuring Input_Data.py, run one of the optimizer scripts directly. In PyCharm, right-click the script and choose Run, or use the terminal.

3.4.1 Using run_sama_optimized.py (recommended for raw-code users)

```
python run_sama_optimized.py --algorithm pso
python run_sama_optimized.py --algorithm ade
python run_sama_optimized.py --algorithm abc
python run_sama_optimized.py --algorithm gwo
```

This script loads sama_config_COMPLETE_HYBRID.yaml (if present) and applies it to InData at runtime, then runs the chosen algorithm.

3.4.2 Running optimizer modules directly

```
python sama/optimizers/pso.py        # Particle Swarm Optimization
python sama/optimizers/ade.py        # Advanced Differential Evolution
python sama/optimizers/run_abc.py    # Artificial Bee Colony
python sama/optimizers/gwo.py        # Grey Wolf Optimizer
```

3.4.3 Using the examples/ folder

The examples/optimization_case.py file contains a complete working example with all 8 electricity rate structures and all 6 natural gas rate structures fully commented. It defines a Config class with all parameters and runs the full optimization workflow.

```
python examples/optimization_case.py
```

4. METHOD 2 - USING THE PYTHON PACKAGE (PIP INSTALL SAMA)

The SAMA pip package (SAMA Py Package) is available through the Python Package Index (PyPI). It provides two console commands, **sama-config** and **sama-run**, that guide you through configuration and run the optimization without editing any source files. This is the recommended approach for most users.

4.1 Installation

7. Install SAMA from PyPI (the SAMA Py Package):
```
pip install sama
```

8. Verify the installation:
```
sama-config --help
sama-run --help
```

Note: Python 3.9 or higher is required. On Windows, ensure Python is in your PATH. All dependencies (numpy, pandas, scipy, numba, matplotlib, openpyxl, seaborn, numpy-financial, questionary, PyYAML) are installed automatically.

4.2 Step 1 - Run the Configuration Wizard

In your working directory (the folder where you want results to be saved), run:
```
sama-config
```

The wizard launches an interactive terminal interface with 18 sections that guide you through every parameter category. You answer questions using your keyboard; arrow keys to select and Enter to confirm. At the end of each section the wizard saves progress and moves to the next.

Wizard Section	Parameters Covered
Section 1: System Components	Which components to include: PV, WT, DG, Battery (lead-acid or Li-ion), Grid, Heat Pump (Bosch or Goodman), EV
Section 2: Optimization Settings	Algorithm (PSO/ADE/ABC/GWO), population size, iterations, run count, design variable bounds (VarMin, VarMax)
Section 3: Project and Calendar	Project lifetime (n), simulation year, public holiday list
Section 4: Electrical Load	Load type selection (1-10), CSV file paths or manual values, previous-year load for NEM calculations
Section 5: Thermal Load	Heat pump thermal load source (house_load.xlsx or manual)
Section 6: Weather and Irradiance	METEO.csv path or manual weather input, PV tilt/azimuth/soiling
Section 7: Constraints	LPSP maximum, renewable energy minimum fraction, budget limit, capacity constraint type

Section 8: Financial	Discount rates, inflation rate, tax rate, RE incentives/federal tax credit
Section 9: PV and Inverter	Pricing method, module efficiency, NOCT parameters, lifetimes, capital/replacement/O&M costs
Section 10: Wind Turbine	Hub height, cut-in/rated/cut-out speeds, power law coefficient, costs
Section 11: Diesel Generator	Rated capacity, fuel curve (a, b), minimum load ratio, emissions, fuel cost, lifetime
Section 12: Battery Storage	Li-ion or lead-acid selection, SOC limits, round-trip efficiency, capacity, lifetime throughput, costs
Section 13: Heat Pump	HP brand (Bosch or Goodman), rated size, lifetime, costs
Section 14: Electric Vehicle	Battery capacity, SOC limits, charging power, range, daily trip, departure/arrival times, degradation
Section 15: Grid and NEM	Net metering flag, capacity option (cap_size), grid tax, annual service charge, sell structure
Section 16: Electricity Rate Structure	Rate type (1-8), prices, TOU hours, seasons, holidays treatment
Section 17: Natural Gas Rate Structure	NG unit (m3/therms/kWh), rate type (1-8), NG prices, escalation
Section 18: Output Settings	Output directory, compare-with-grid scenario flag

When the wizard completes, it saves a file called sama_config_COMPLETE.yaml in your working directory. This YAML file contains all your parameters and can be edited manually or re-run through the wizard at any time.

4.3 Step 2 - Run the Optimization

With sama_config_COMPLETE.yaml in your working directory, run:

```
sama-run
```

SAMA will automatically find the config file, load all parameters, and run the optimization algorithm specified in the config (optimization_algorithm key). Additional options are available:

4.3.1 sama-run command options

Option	Description	Example
-c / --config	Path to config YAML (default: sama_config_COMPLETE.yaml in cwd)	sama-run -c my_project.yaml
-a / --algorithm	Override the algorithm from the config file	sama-run -a pso
--output	Override the output directory from the config	sama-run --output results/run1

--dry-run	Validate the config file without running optimization	sama-run --dry-run --verbose
--no-gui	Disable matplotlib GUI (use Agg backend for servers)	sama-run --no-gui
--verbose	Show full parameter log and config summary	sama-run --verbose

The optimization_algorithm key in the YAML accepts: pso, ade, abc, or gwo.

4.4 Step 3 - View Results

Results are saved to the output directory specified in the config (default: sama_outputs/ in your working directory). The folder is organized as:

```
sama_outputs/
    Optimization.png    convergence curve (best cost vs. iteration)
    figs/               all chart PNG/SVG files
    data/               CSV data files
```

4.4.1 Chart files generated in sama_outputs/figs/

File	Description
Cash_Flow.svg	25-year cash flow breakdown (capital, replacement, O&M, fuel, grid)
Cash Flow_ADV.png	Advanced cash flow with grid comparison (when compare_with_grid = 1)
Multiple_Cash_Flow_ADV.png	Multi-run cash flow comparison (when Run_Time > 1)
Energy Distribution.png	Annual energy supply breakdown by source (PV, WT, DG, battery, grid)
Battery State of Charge.png	Hourly battery SOC profile over the year
Grid Interconnection.png	Hourly grid import and export over the year
Specific day results.png	Detailed 24-hour power flow for a representative day
Daily-Monthly-Yearly average cost of energy system.png	Cost comparison across daily, monthly, and annual timescales
Daily-Monthly-Yearly average hourly cost of connecting to the grid.png	Hourly grid cost patterns by season
Daily-Monthly-Yearly average cost of only grid-connected system.png	Baseline grid-only cost for comparison
Grid Hourly Cost.png	Hourly grid electricity cost profile
Daily-Monthly-Yearly average earning Sell to the Grid.png	Revenue from grid exports (if NEM enabled)

Energy Balance.png	Hourly energy balance (supply vs. demand)
electricity_comparison.png	SAMA system vs. grid-only electricity cost comparison
EV Energy.png	EV charging and discharging energy over the year
EV Sp Results 1.png	EV-specific results chart 1 (when EV = 1)
EV Sp Results 2.png	EV-specific results chart 2 (when EV = 1)
hp_ambient_conditions.png	Ambient temperature and pressure conditions for heat pump (when HP = 1)
hp_heating_performance.png	Heat pump heating power and COP (when HP = 1)
hp_cooling_performance.png	Heat pump cooling power and COP (when HP = 1)
hp_cop_vs_temp.png	COP as a function of ambient temperature (when HP = 1)
hp_monthly_summary.png	Monthly heat pump energy and COP summary (when HP = 1)
Optimization.png	Optimizer convergence curve (saved in sama_outputs/ root)

4.4.2 Data files generated in sama_outputs/data/

File	Description
Outputforplotting.csv	Complete hourly time-series: PV output, WT output, battery SOC, DG power, grid import/export, load, EV charge/discharge, and all cost variables for all 8,760 hours
cash_flow_data.csv	Annual cash flow breakdown over the 25-year project lifetime

4.5 Using the Python API

After pip install sama, SAMA can also be used directly in Python scripts without the command-line interface:

4.5.1 Basic API workflow

```
from sama.core.Input_Data import InData
from sama.cli.config_loader import load_config, apply_config

# Load a YAML config produced by sama-config
config = load_config('sama_config_COMPLETE.yaml')

# Apply all parameters to the InData singleton
apply_config(config)

# Run the optimizer
from sama.optimizers.swarm import Swarm
swarm = Swarm()
swarm.optimize()
```

4.5.2 Available optimizer classes

Algorithm	Import	Class
Particle Swarm Optimization	from sama.optimizers.swarm import Swarm	Swarm()
Advanced Differential Evolution	from sama.optimizers.AdvancedDifferentialEvolution import AdvancedDifferentialEvolution	AdvancedDifferentialEvolution()
Artificial Bee Colony	from sama.optimizers.ArtificialBeeColony import ImprovedArtificialBeeColony	ImprovedArtificialBeeColony()
Grey Wolf Optimizer	from sama.optimizers.GreyWolfOptimizer import GreyWolfOptimizer	GreyWolfOptimizer()

4.5.3 Config loader helper functions

Function	Description
load_config(path)	Load a YAML config file and return it as a Python dictionary
apply_config(config)	Apply a config dictionary to the InData singleton and return the fully reconfigured InData instance
save_config(config, path)	Save a config dictionary to a YAML file
merge_configs(*paths)	Merge multiple YAML config files into one dictionary (later files override earlier ones)

4.5.4 Important: Package Name vs Import Name

The SAMA Python package is published on PyPI under the distribution name SAMAbyRenXera. This is the name you use with pip to install the package. However, once installed, the package is imported in Python code using the name sama (all lowercase). These are two different names for the same package:
Install with pip (use the PyPI distribution name):

```
pip install SAMAbyRenXera
```

Import in Python scripts (use the internal package name):

```
import sama
```

Command line tools installed automatically:

```
sama-config
sama-run
```

When searching for the package on the PyPI website, look for SAMAbyRenXera. When writing Python scripts that use SAMA, always write import sama.

4.6 Using Your Own Custom Data Files

SAMA ships with bundled default data files for London, Ontario (weather, electrical load, and heat pump data). When you run sama-config and provide your own file paths, the wizard automatically copies those files into the local sama/content/ folder in your working directory. SAMA reads from this local folder instead of the bundled defaults.

You can also copy your custom files directly into the sama/content/ folder yourself, and SAMA will use them automatically when you run sama-run. The sama/content/ folder is created in your working directory the first time you run sama-config or sama-run.
The expected file names and formats are:
Eload.csv: Hourly electrical load profile. One column, 8,760 rows, no header. Values in kW.
METEO.csv: SAM NSRDB weather file for your location. Two header rows followed by 8,760 hourly data rows. Must include columns: GHI, DNI, DHI, Temperature, Wind Speed, Pressure. Download from https://nsrdb.nrel.gov/
house_load.xlsx: Building thermal load. Excel file with heating load (Hload) in column 2 and cooling load (Cload) in column 3, 8,760 rows each, in kW.
Irradiance.csv: Only needed when G_type = 2 (manual plane-of-array irradiance). One column, 8,760 rows, no header. Values in W/m2.
If a file is not present in the local sama/content/ folder, SAMA automatically falls back to the bundled default. You only need to provide files you want to customize.

4.7 Typical Workflow for a New Project Location

Step 1: Create an empty project folder, for example C:\Projects\MyCity_SAMA
Step 2: Download your METEO.csv weather file from https://nsrdb.nrel.gov/ for your project location.
Step 3: Prepare your Eload.csv file with 8,760 rows of hourly electrical load data in kW.
Step 4: If modeling a heat pump, prepare house_load.xlsx with Hload and Cload columns.
Step 5: Open a terminal in your project folder and run sama-config. The wizard will ask for paths to your data files and copy them into sama/content/ automatically.
Step 6: When the wizard finishes, sama_config_COMPLETE.yaml is saved in your project folder. You can open it in any text editor to review or manually adjust settings.
Step 7: Run sama-run to start the optimization. Results are saved in sama_outputs/ inside your project folder.

4.8 Understanding the Working Directory

SAMA creates all its folders (sama/content/, sama_inputs/, sama_outputs/) relative to the folder where you run the sama-config and sama-run commands. This folder is called the working directory. Always run both commands from the same project folder so that the configuration file, custom data files, and results all stay together in one place.
On Windows: navigate to your project folder in File Explorer, click the address bar, type cmd, and press Enter. This opens a Command Prompt already pointed at that folder.

4.9 What to Expect on the First Run

The very first time you run **sama-run** after installation, SAMA uses Numba to compile the energy management system (EMS) functions. This compilation is a one-time process and can take several minutes. You will see messages in the terminal while this happens. Subsequent runs will be significantly faster because the compiled code is cached on your computer.
For a typical run with MaxIt=200 and nPop=50, expect the optimization itself to take between 5 and 30 minutes depending on your hardware and which components (EV, heat pump) are enabled.

5. METHOD 3 - WINDOWS .EXE APPLICATION (ALPHA)

A Windows executable (.exe) version of SAMA is currently in the alpha development phase and is available on request from the developers. This version provides a graphical user interface (GUI) that does

not require Python to be installed on the user's computer. It is intended for non-technical users who prefer a point-and-click experience.

Because this version is in the alpha phase, some features may be incomplete or subject to change. Users are encouraged to report any issues to the development team.

To request the Windows .exe version, contact the FAST research group at Western University. Contact information is available at the GitHub repository: https://github.com/Sas1997/SAMA

Windows .exe Application (Alpha Development Phase):
The .exe version is not distributed via PyPI.
It must be obtained directly from the developers.
No Python installation is required to run the .exe.
It uses the same algorithms and produces the same outputs as the Python package.
This version is currently in the alpha development phase. Features may be incomplete.

6. THE CONFIGURATION FILE (SAMA_CONFIG_COMPLETE.YAML)

Whether you use the wizard (sama-config) or edit it manually, the configuration file is a standard YAML text file. You can open it in any text editor. Below is a complete reference of all keys.

6.1 File Structure Overview

```
input_directory: C:\path\to\project          # working directory (used by
raw code)
output_directory: C:\path\to\sama_outputs   # results output folder
optimization_algorithm: abc                # pso | ade | abc | gwo

# Optimization settings
MaxIt: 200
nPop: 50
Run_Time: 1
VarMin: [0.0, 0.0, 0.0, 0.0, 0.0]
VarMax: [60.0, 60.0, 60.0, 20.0, 60.0]

# System components (1=included, 0=excluded)
PV: 1   WT: 1   DG: 1   Bat: 1   Grid: 1   HP: 1   EV: 1
Li_ion: 1   Lead_acid: 0
NEM: 1

# Project calendar
n: 25           # project lifetime (years)
year: 2023      # simulation year
holidays: [1, 51, 97, ...]   # day-of-year numbers for public holidays
```

6.2 Key Parameter Reference

YAML key	Type	Default	Description
Optimization			
MaxIt	int	200	Maximum optimizer iterations
nPop	int	50	Population size (number of candidate solutions)
Run_Time	int	1	Independent optimization runs (best result kept)
VarMin / VarMax	list[4]	[0,0,0,0,0] / [60,60,60,20,60]	Lower/upper bounds for [Npv, Nwt, Nbat, N_DG, Cn_I]
Cash_Flow_adv	int	0	0 = standard cash flow; 1 = advanced multi-run cash flow chart
Calendar			
n	int	25	Project lifetime in years

year	int	2023	Simulation year (affects day-of-week and leap year)
holidays	list[int]	[1,51,97,...]	Day-of-year numbers for public holidays (affects TOU pricing)
Load			
load_type	int	1	Electrical load input mode (1-10, see Section 3.3.3)
load_previous_year_type	int	1	Previous year load mode for NEM credit calculation (1-11)
Tload_type	int	1	Thermal load input type (1 = from house_load.xlsx)
Weather			
G_type	int	1	Irradiance input mode (1=NSRDB/SAM, 2=CSV)
T_type	int	1	Temperature input mode (1=NSRDB/SAM, 2=CSV, 3=monthly avg, 4=annual avg)
WS_type	int	1	Wind speed input mode (1=NSRDB/SAM, 2=CSV, 3=monthly avg, 4=annual avg)
azimuth	float	180.0	PV panel azimuth angle (degrees, 180 = south-facing)
tilt	float	33.0	PV panel tilt angle (degrees)
soiling	float	5.0	PV soiling losses (%)
weather_url	str	METEO.csv	Path to NSRDB/SAM meteorological CSV file
Constraints			
LPSP_max_rate	float	0.000999	Maximum LPSP (% of annual demand that can be unmet)
RE_min_rate	float	50.0	Minimum renewable energy fraction required (%)
EM	int	0	0 = minimize NPC; 1 = minimize NPC + LEM (emissions)
Budget	float	2,000,000	Maximum allowable capital cost ($)
cap_option	int	1	1=size cap (kW), 2=annual load, 3=rooftop area, 4=no limit
cap_size	float	15.0	PV system size cap in kW (used when cap_option=1)
Financial			

n_ir_rate	float	2.75	Nominal discount rate (%)
e_ir_rate	float	2.0	Expected inflation rate (%)
Tax_rate	float	0.0	Equipment sales tax (%)
RE_incentives_rate	float	30.0	Federal/provincial RE incentives / tax credit (%)
Pricing_method	int	2	1=top-down (Total_PV_price $/kW), 2=bottom-up (itemized)
Grid			
Grid_sale_tax_rate	float	13.0	Electricity sales tax (%)
Grid_Tax_amount	float	0.0353	Fixed grid tax in $/kWh (e.g. delivery charge component)
Grid_credit	float	0.0	Annual grid credit to customers in $ (e.g. rebate)
Grid_escalation_rate	list[12]	2.0 per year	Annual electricity price escalation rate per year (%)
NEM	int	1	1 = net metering enabled (annual credit reconciliation)
NEM_fee	float	0.0	One-time NEM interconnection setup fee ($)
Monthly_fixed_charge_system	int	1	1=flat service charge, 2=tiered service charge
SC_flat	float	27.16	Flat monthly service charge ($)
rateStructure	int	7	Electricity rate type (1-8, see Section 3.3.7)
sellStructure	int	1	Grid sell price structure (1=flat, 2=monthly, 3=same as buy)
Pbuy_max	float	50.0	Maximum grid import power (kW)
Psell_max	float	50.0	Maximum grid export power (kW)
Natural Gas			
NG_energycontent	float	10.81	Energy content of 1 m3 of natural gas (kWh/m3)
Furnace_eff	float	0.94	Efficiency of natural gas furnace (fraction)
NG_unit	str	therms	Unit of NG prices entered: m3, therms, or kwh
rateStructure_NG	int	1	NG rate type (1-8)
SC_flat_NG	float	18.59	Flat monthly NG service charge ($)

7. OPTIMIZATION ALGORITHMS

SAMA includes four metaheuristic optimization algorithms. All four optimize the same objective (NPC, or NPC+LEM when EM=1) over the same 5-variable design space. The choice of algorithm does not affect what is being optimized, only how the search space is explored.

7.1 Particle Swarm Optimization (PSO)

PSO simulates a swarm of candidate solutions (particles) that move through the search space. Each particle adjusts its velocity based on its own best-known position and the swarm's global best position. Key parameters:

Parameter	Default	Description
w	1.0	Inertia weight: controls how much a particle keeps its current velocity
wdamp	0.99	Inertia weight damping ratio: w is multiplied by wdamp each iteration
c1	2.0	Personal learning coefficient: attraction toward particle's own best position
c2	2.0	Global learning coefficient: attraction toward swarm's global best position

7.2 Advanced Differential Evolution (ADE)

ADE is a population-based algorithm that creates new candidate solutions by combining existing ones using differential mutation and crossover. SAMA's ADE implementation uses multiple DE strategies with adaptive parameters:

- Five mutation strategies: DE/rand/1, DE/best/1, DE/current-to-best/1, DE/rand/2, DE/best/2
- Adaptive scaling factor F and crossover probability CR, tuned by success rate and iteration progress
- Heterogeneous initialization (random, center-biased, boundary-biased, Sobol-like) for diverse coverage

Parameter	Default	Description
F_min / F_max	0.1 / 0.9	Minimum/maximum scaling factor for mutation
CR_min / CR_max	0.1 / 0.9	Minimum/maximum crossover probability

7.3 Improved Artificial Bee Colony (ABC)

ABC mimics the foraging behavior of honey bees. It has three phases: employed bees, onlooker bees, and scout bees, each exploring and exploiting promising regions of the search space. SAMA's ABC uses multi-dimensional perturbations and guided reinitialization of stagnant solutions.

Parameter	Default	Description
maxTrials	15	Maximum number of trials before a food source is abandoned and re-scouted

modification_rate	0.8	Probability of modifying multiple dimensions simultaneously
initial_search_radius	0.5	Search radius at the start of optimization (fraction of variable range)
final_search_radius	0.1	Search radius at the end of optimization

7.4 Grey Wolf Optimizer (GWO)

GWO mimics the social hierarchy and hunting behavior of grey wolves. Solutions are classified as alpha (best), beta, delta, and omega wolves. The pack converges on prey (optimal solution) guided by alpha, beta, and delta wolves. The parallel variant (ParallelGreyWolfOptimizer) uses Python multiprocessing for faster evaluation when Run_Time > 1.

7.5 Choosing an Algorithm

All four algorithms are suitable for this problem. In practice, for the default 5-variable search space with MaxIt=200 and nPop=50, any algorithm will find good solutions. ADE and ABC tend to explore more thoroughly; PSO and GWO converge faster. For a rigorous comparison, use Run_Time > 1 to average across multiple independent runs.

8. ENERGY MANAGEMENT SYSTEM (EMS)

For each candidate system design, SAMA simulates 8,760 hours of operation using an hourly dispatch model. The EMS decides, for every hour, how much power each component supplies or consumes.

8.1 Base Dispatch (EMS.py)

The base EMS uses Numba JIT compilation for fast execution. In each hour it performs marginal-cost-based dispatch:

9. Calculate available PV and wind power from irradiance and wind speed.
10. Calculate net demand (load minus PV minus wind output).
11. If net demand is positive (deficit): draw from battery, then diesel generator, then grid, based on relative hourly cost.
12. If net demand is negative (surplus): charge battery, then export to grid if NEM is enabled.
13. Track battery SOC, respect SOC_min, SOC_max, and charge/discharge current limits.
14. Track any unmet energy (Ens) toward LPSP calculation.

8.2 EV Smart Charging and V2X (EMS_EV.py)

When EV = 1, the EMS_EV module adds vehicle-to-home (V2H) and vehicle-to-grid (V2G) capabilities. Key behaviors:

- EV is only available to charge or discharge when it is at home (EV_p = 1, determined by departure time Tout, arrival time Tin, and holiday schedule).
- Departure planning: the EV must reach SOC_dep before Tout each day.
- Lookahead arbitrage: SAMA looks ahead at future grid prices to decide whether to discharge the EV battery now (selling to grid or powering the home) or wait. The decision accounts for battery wear cost and round-trip efficiency losses.
- Safety buffer: a reserve SOC is maintained based on historical departure patterns and load variability.
- V2H discharge is used when it is economically advantageous (current buy price greater than battery wear cost plus discharge losses).

Key EV parameters set in Input_Data.py or the wizard:

Parameter	Default	Description
C_ev	82 kWh	EV battery total capacity
SOCe_min / SOCe_max	0.03 / 0.97	Minimum and maximum usable SOC fraction
Pev_max	11 kW	Maximum charge and discharge power
Range_EV	468 km	EV full-charge driving range
Daily_trip	68 km	Average daily distance driven
SOC_dep	0.85	Required SOC at departure time
Tin / Tout	17 / 8	Arrival hour (17 = 5 PM) and departure hour (8 = 8 AM)

n_e	0.9	EV battery round-trip charge efficiency
L_EV_dis	400,000 km	EV battery lifetime in total km driven
treat_special_days_as_home	false	If true, holidays are treated as home days for EV presence

8.3 Heat Pump Energy Attribution (EMS_HP.py)

When HP = 1, EMS_HP post-processes the dispatch results to attribute the electricity consumed by the heat pump to heating versus cooling loads. Two methods are used: proportional allocation (based on hourly heating and cooling fractions) and marginal-cost-based assignment. This enables detailed multi-fuel cost accounting that separates electricity costs for space heating from space cooling.

9. COMPONENT MODELS

9.1 Solar PV

Hourly PV power output uses a temperature-dependent efficiency model with NOCT-based cell temperature correction:

- Cell temperature is calculated from ambient temperature, NOCT, and irradiance.
- Module efficiency is adjusted for cell temperature using Tcof (temperature coefficient, default -0.3 %/C).
- Output is multiplied by the derating factor fpv (default 0.9) for real-world losses.
- The default module rated power is Ppv_r = 1 kW; the optimizer returns the number of modules Npv.

9.2 Wind Turbine

- Hub-height wind speed is calculated from anemometer height h0 to hub height h_hub using a power-law profile with friction coefficient alfa_wind_turbine (default 0.14).
- Power output uses a piecewise model: zero below cut-in speed (v_cut_in = 2.5 m/s), linear ramp to rated power between cut-in and rated speed (v_rated = 9.5 m/s), rated power between rated and cut-out (v_cut_out = 25 m/s), and zero above cut-out.
- The default rated power is Pwt_r = 1 kW; the optimizer returns the number of turbines Nwt.

9.3 Battery Storage

SAMA supports two battery chemistries:

- Li-ion: idealized model with fixed round-trip efficiency ef_bat_Li (default 0.90), maximum charge current Ich_max_Li_ion, maximum discharge current Idch_max_Li_ion, lifetime throughput Q_lifetime_Li (default 3,000 kWh per unit), and nominal capacity Cnom_Li.
- Lead-acid: KiBaM (Kinetic Battery Model) with capacity ratio c and storage rate constant k_lead_acid, maximum charge current Ich_max_leadacid, and lifetime throughput Q_lifetime_leadacid (default 8,000 kWh per unit).

Both types use SOC_min (default 0.1) and SOC_max (default 1.0) to bound the state of charge. The battery rated capacity Cbt_r is derived from the cell voltage and nominal capacity.

9.4 Diesel Generator

Fuel consumption is modeled as an affine function of output power: F = a * Prated + b * P_output (L/hr). Parameters a = 0.4388 and b = 0.1097 are default values. The minimum loading ratio LR_DG (default 0.25) prevents the generator from running at very low fractions of rated capacity. Emissions per litre of fuel are set by CO2, CO, NOx, and SO2.

9.5 Heat Pump Models

SAMA includes black-box models for two heat pump brands:

- Goodman (BB_HP_Goodman.py): Uses manufacturer data to compute hourly electric consumption for space heating and cooling as a function of outdoor ambient temperature and building thermal load (Hload, Cload). Returns COP curves for heating and cooling modes.
- Bosch (BB_HP_Bosch.py): Same structure as the Goodman model but using Bosch manufacturer data. Select using HP_brand = 'Bosch' or HP_brand = 'Goodman'.

Both models read lookup data from the content/ folder (HP_Bosch/ and HP_Goodman/ subdirectories containing Excel data files).

10. FINANCIAL MODEL

10.1 Net Present Cost (NPC)

NPC is the total 25-year lifecycle cost of the hybrid energy system in today's dollars. It includes:

- Initial capital cost (I_Cost): PV, WT, DG, battery, inverter, charger, engineering costs, heat pump, EV battery, NEM connection fee, reduced by RE incentives and multiplied by (1 + System_Tax).
- Replacement costs (R_Cost): scheduled component replacements over the project lifetime, discounted to present value.
- O&M costs (MO_Cost): annual operations and maintenance costs for all components, escalated and discounted.
- Fuel cost (C_Fu): diesel generator fuel cost, escalated by C_fuel_adj_rate and discounted.
- Grid electricity cost (Grid_Cost_net): annual grid purchases at tariff rates, escalated by Grid_escalation_rate and discounted. NEM credits and grid credits are subtracted.
- Salvage value: remaining value of components at end of project life.

NPC = (I_Cost + sum(R_Cost) + sum(MO_Cost) + sum(C_Fu) - sum(Salvage)) * (1 + System_Tax) + sum(Grid_Cost_net) + sum(gHeating_Cost)

10.2 Levelized Cost of Energy (LCOE)

LCOE = CRF * NPC / E_tot, where CRF is the capital recovery factor derived from the real discount rate ir and project lifetime n, and E_tot is the total discounted energy supplied over the project lifetime (including EV charging and thermal energy from the heat pump).

10.3 Levelized Emissions (LEM)

LEM is computed when EM = 1. It is the total lifecycle emissions (from diesel generation and grid-sourced electricity) normalized by total supplied energy. Emission factors CO2, CO, NOx, SO2 (kg/L of diesel) and E_CO2, E_SO2, E_NOx (kg/kWh of grid electricity) are user-specified.

10.4 25-Year Cash Flow Projection

The financial model projects cash flows year by year over the n-year project lifetime. Grid electricity prices are escalated using Grid_escalation_rate (array of 25 annual percentages). Natural gas prices are escalated using Grid_escalation_rate_NG. Component replacements are scheduled at multiples of their lifetimes (L_PV, L_WT, L_B, L_I, L_CH, L_HP, L_EV). The Advanced_multi_cashflow module generates multi-run comparison charts when Cash_Flow_adv = 1 and Run_Time > 1.

11. BUNDLED DATA FILES (CONTENT/ FOLDER)

The following data files are bundled with the SAMA package in the sama/content/ directory and can be accessed via get_content_path():

File	Description
Eload.csv	Sample hourly electrical load profile (8,760 rows, kW) for a representative building in London, Ontario
Eload_daily.csv	Sample daily electrical load profile (24 rows) for generic load scaling
METEO.csv	SAM NSRDB meteorological data for London, Ontario. Columns include GHI, DNI, DHI, temperature, wind speed, and pressure
Irradiance.csv	Sample hourly plane-of-array (POA) irradiance CSV (8,760 rows, W/m2) for manual G_type=2 input
Temperature.csv	Sample hourly ambient temperature CSV (8,760 rows, C) for manual T_type=2 input
WSPEED.csv	Sample hourly wind speed CSV (8,760 rows, m/s) for manual WS_type=2 input
Battery.csv	Battery lookup data
Data.csv	General component reference data
Generic_load_JulyP.csv	Normalized generic hourly load profile with July peak; used when load_type = 5/8/9/10
Generic_load_JanuaryP.csv	Normalized generic hourly load profile with January peak; used when load_type = 5/8/9/10
house_load.xlsx	Sample building heating (Hload) and cooling (Cload) thermal load (8,760 rows, kW each) for London, Ontario
HP_Bosch/	Bosch heat pump manufacturer performance data files (6 Excel files and weather data)
HP_Goodman/	Goodman heat pump manufacturer performance data files (8 Excel files and weather data)

To use your own weather or load data, either replace these files in place or specify alternative file paths in Input_Data.py (e.g. path_Eload, path_G, path_T, path_WS, weather_url) or in the config YAML.

12. INPUT DATA FORMATS

12.1 Electrical Load CSV (Eload.csv)

- One column, 8,760 rows (no header).
- Each row is the hourly average electrical load in kW for that hour of the year.
- Hour 0 = midnight on January 1; hour 8,759 = 11 PM on December 31.

12.2 METEO.csv (SAM NSRDB format)

- Two header rows followed by hourly data rows (8,760 rows).
- Must include columns: GHI, DNI, DHI, Temperature, Wind Speed, Pressure.

- Downloaded from NSRDB (https://nsrdb.nrel.gov/) or exported from SAM (System Advisor Model).

12.3 Plane-of-Array Irradiance CSV (Irradiance.csv)

- One column, 8,760 rows (no header).
- Each row is the hourly POA irradiance in W/m2.
- Used when G_type = 2.

12.4 house_load.xlsx (Thermal Load)

- Excel file with at least 3 columns.
- Column index 1 (second column): hourly heating load Hload in kW (8,760 rows).
- Column index 2 (third column): hourly cooling load Cload in kW (8,760 rows).

13. TROUBLESHOOTING

Error / Symptom	Cause	Fix
ModuleNotFoundError: No module named 'sama'	SAMA not installed in current Python environment	Run pip install sama (or pip install -e . from project root)
command not found: sama-config	pip install did not add scripts to PATH	On Windows: restart terminal. On Linux/Mac: add ~/.local/bin to PATH
BackendUnavailable: Cannot import setuptools.backends.legacy	Old version of setuptools	The pyproject.toml uses setuptools.build_meta. Ensure the correct pyproject.toml is present.
ModuleNotFoundError: No module named 'numba'	numba not installed or wrong version for Python 3.13+	pip install 'numba>=0.60'
_tkinter.TclError: no display name	matplotlib trying to open a GUI window on a headless server	Use sama-run --no-gui, or set environment variable MPLBACKEND=Agg
FileNotFoundError: Eload.csv not found	Package data not installed correctly	Run pip install --force-reinstall sama
Optimization produces 0 kW for all components	All candidate solutions violate constraints (LPSP, RE_min, Budget)	Relax constraints: increase LPSP_max_rate, decrease RE_min_rate, increase Budget or VarMax
sama-config freezes or no text appears	Terminal does not support questionary interactive prompts	Try a different terminal: cmd.exe or PowerShell on Windows; bash or zsh on Linux/macOS
Very slow optimization (many hours)	numba JIT compilation on first run, or MaxIt/nPop too large	First run includes JIT compilation (expected). Reduce MaxIt or nPop for testing.
Results folder is empty after optimization	Output directory not created or permissions issue	Check output_directory in YAML config. Ensure write permissions in that folder.

14. QUICK REFERENCE - MINIMUM WORKING EXAMPLE

The following is the minimum set of steps to run SAMA with a fresh pip installation using default bundled data:

a) Install SAMA (the SAMA Py Package):

```
pip install sama
```

b) Open a terminal in an empty working directory.

c) Run the configuration wizard:

```
sama-config
```

Follow all 18 sections. For a quick test, accept defaults in most sections.

d) Run the optimization:

```
sama-run
```

e) View results in sama_outputs/figs/ and sama_outputs/data/Outputforplotting.csv.

Expected Runtime:
The first run always takes longer because numba compiles the EMS functions on first call.
After the first run, subsequent runs are significantly faster.
A typical run with MaxIt=200, nPop=50 takes 5-30 minutes depending on hardware and which components are enabled.

15. PACKAGE MODULE STRUCTURE

Module / File	Description
sama/core/	
Input_Data.py	Main parameter class: defines all input data and default values. Singleton pattern: InData = Input_Data()
Fitness.py	Fitness evaluation function: calls EMS for each candidate design and returns NPC, LCOE, LEM, LPSP, and constraint penalties
sama/cli/	
wizard.py	Entry point for sama-config command: launches sama_Wizard.py
sama_Wizard.py	Interactive 18-section configuration wizard using questionary
runner.py	Entry point for sama-run command: loads YAML, applies config, runs selected algorithm
config_loader.py	load_config(), apply_config(), save_config(), merge_configs() functions
sama/optimizers/	
swarm.py	Particle Swarm Optimization (Swarm class)
AdvancedDifferentialEvolution.py	Advanced Differential Evolution (AdvancedDifferentialEvolution class)
ArtificialBeeColony.py	Improved Artificial Bee Colony (ImprovedArtificialBeeColony class)
GreyWolfOptimizer.py	Grey Wolf Optimizer (GreyWolfOptimizer and ParallelGreyWolfOptimizer classes)
pso.py / ade.py / run_abc.py / gwo.py	Thin wrapper scripts for running each algorithm directly from command line
sama/ems/	
EMS.py	Base hourly dispatch engine (Numba JIT compiled)
EMS_EV.py	Extended dispatch engine with EV smart charging and V2X lookahead arbitrage
EMS_HP.py	Heat pump energy attribution post-processor
sama/models/	
Battery_Model.py	Battery discharge and degradation models (lead-acid KiBaM and Li-ion)
BB_HP_Goodman.py	Goodman air-source heat pump black-box model
BB_HP_Bosch.py	Bosch air-source heat pump black-box model
sama/pricing/	
Electricity_Bill_Calculator.py	Monthly and annual billing calculations with escalation and NEM reconciliation

calcFlatRate.py	Flat rate tariff calculator
calcSeasonalRate.py	Seasonal rate tariff calculator
calcMonthlyRate.py	Monthly rate tariff calculator
calcTieredRate.py	Tiered rate tariff calculator
calcSeasonalTieredRate.py	Seasonal tiered rate calculator
calcMonthlyTieredRate.py	Monthly tiered rate calculator (3-tier)
calcMonthlyTieredRate4.py	Monthly tiered rate calculator (4-tier, for Enbridge EGD)
calcTouRate.py	Time-of-Use (TOU) rate calculator
calcULTouRate.py	Ultra-Low TOU (ULO) rate calculator
service_charge.py	Monthly utility service charge calculator (flat or tiered)
Advanced_multi_cashflow.py	Advanced multi-run cash flow analysis and comparison plots
sama/results/	
Results.py	Gen_Results(): generates all output charts and the Outputforplotting.csv data file
sama/utilities/	
daysInMonth.py	Calendar utility: leap-year aware days-per-month array
dataextender.py	Expand monthly averages to 8,760-hour arrays
generic_load.py	Generic electrical load profile generator (July-peak and January-peak archetypes)
sam_monofacial_poa.py	SAM NSRDB API integration: plane-of-array irradiance and temperature from METEO.csv
top_down.py	Top-down PV pricing calculator
EV_Presence.py	Determine hourly EV home-presence schedule from departure/arrival times and holidays
EV_travel.py	EV travel energy consumption modeling
EV_demand_dest.py	EV demand and destination distribution utilities
Ev_Battery_Throughput.py	EV battery lifetime throughput calculation based on degradation and km driven
sama/content/	
(data files)	Bundled CSV, Excel, and meteorological data files (see Section 11)

16. LICENSE AND CITATION

16.1 License

SAMA is released under the GNU General Public License v3.0 (GPL-3.0). You are free to use, copy, modify, and distribute SAMA under the terms of this license. Any derivative works must also be released under GPL-3.0.

16.2 Authors

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16.3 Citing SAMA

If you use SAMA in academic work, please cite the primary published paper [A1]. A Journal of Open Source Software (JOSS) paper is currently pending submission; once published, that citation should also be included. In the meantime, please cite the GitHub repository and acknowledge the FAST Research Group at Western University.

Primary citation (published):

Sadat, S.A., Takahashi, J. and Pearce, J.M., 2023. A Free and open source microgrid optimization tool: SAMA the solar alone Multi-Objective Advisor. Energy Conversion and Management, 298, p.117686. https://doi.org/10.1016/j.enconman.2023.117686

JOSS paper (pending submission):

Sadat, S.A. and Pearce, J.M. SAMA: Solar Alone Multi-Objective Advisor. Journal of Open Source Software (JOSS). Pending submission.

Additional SAMA publications (see Section 19 for full list):
Sadat & Pearce (2024). Economic grid defection analysis. Solar Energy, 282, 112910.
Sadat & Pearce (2025). Electricity pricing structures in Canada. Renewable Energy, 242, 122456.
Repository: https://github.com/Sas1997/SAMA | FAST Research Group, Western University | GPL-3.0 License

17. Technical Framework and Mathematical Formulation

This section presents the mathematical models and equations that underpin SAMA's simulation and optimization engine. The framework described here is drawn from the original SAMA technical documentation [A1] and peer-reviewed literature. All equation numbers correspond to the formulations implemented in the source code.

17.1 Particle Swarm Optimization Algorithm

The default optimizer in SAMA is the Particle Swarm Optimizer (PSO), a bio-inspired algorithm that moves a population of candidate solutions (particles) through the search space according to the following velocity and position update equations [A1]:

$$V_i^{t+1} = W.V_i^t + c_1 U_1^t\left(P_{b_1} - P_i^t\right) + c_2 U_2^t(g_b^t - P_i^t) \quad (1)$$

$$P_i^{t+1} = P_i^t + v_i^{t+1} \quad (2)$$

In Eq. 1, W is the inertia weight, c1 is the cognitive (personal learning) constant, c2 is the social (global learning) constant, U1 and U2 are random numbers drawn uniformly from [0,1], Pb is the particle's personal best position, and gb is the global best position of the swarm. Eq. 2 updates the particle position using the new velocity. Default PSO parameters in SAMA are W = 1, c1 = 2, c2 = 2, with an inertia weight damping ratio of 0.99 per iteration.

17.2 Solar PV Power Output Model

The hourly power output of PV modules (P_PV [kW]) and corresponding PV energy (E_PV [kWh]) are functions of module temperature (T_Module [degC]) and plane-of-array irradiance (POA [W/m2]), calculated as follows [13][14][15]:

$$P_{PV}\ [kW] = N_{PV} \times f_{PV} \times P_{PV}^{STC} \times \left(\frac{POA}{POA_{STC}}\right)\left[1 + \delta_{PV}\left(T_{Module} - T_{\mathrm{Ref}}\right)\right] E_{PV} = P_{PV} \times t \quad (3)$$

Where N_PV is the optimum number of PV modules (from optimizer), f_PV is the PV derating factor (default 0.9) [7], P_PV_STC [kW] is the rated capacity at Standard Test Conditions (STC: 1000 W/m2 irradiance, 25 degC), POA_STC = 1000 W/m2 is the reference irradiance [10], delta_PV is the temperature coefficient at STC (default -3.7 x 10^-3 /degC) [16], and T_Ref = 25 degC is the STC reference temperature [8].
The module operating temperature T_Module is calculated using the Nominal Operating Cell Temperature (NOCT) model [17]:

$$T_{Module}[°C] = T_{Amb} + \left(\frac{T_{noct} - 20}{800}\right) \times POA \quad (4)$$

Where T_noct [degC] is the nominal operating cell temperature (default 45 degC) [9] and T_Amb [degC] is the ambient temperature from the meteorological dataset.

17.3 Battery Storage Model (KiBaM)

SAMA uses the Kinetic Battery Model (KiBaM) [18] to model energy storage, which was originally developed for lead-acid batteries and has been validated for lithium-ion and other chemistries [19][20][21][22]. KiBaM represents the battery as two reservoirs (available and bound energy). The maximum chargeable power P_BT_ch_max_KiBaM [kW] is given by [18]:

$$P_{BT,ch}^{\mathrm{max,KiBaM}} = \frac{kcQ_{Max} + kQ_1 e^{-kt} + Qkc(1 - e^{-kt})}{1 - e^{-kt} + c(k\Delta t - 1 + e^{-kt})}\ ; E_{BT,ch}^{\mathrm{max,KiBaM}} = P_{BT,ch}^{\mathrm{max,KiBaM}} \times t \quad (5)$$

Where Q1 is the available energy at the start of the time-step [kWh], Q_max is the total storage capacity [kWh], Q is total energy in storage at the start of the time-step [kWh], c is the storage capacity ratio, k is the storage rate constant [h^-1], and t is the simulation time-step. Following HOMER Pro [23], SAMA also applies two additional charging limits based on the maximum charge rate (alpha [A/Ah]) and maximum current (I_max):

$$P_{BT,ch}^{\mathrm{max,mcr}} = \frac{(1 - e^{-\alpha t})(Q_{max} - Q)}{t} E_{BT,ch}^{\mathrm{max,mcr}} = P_{BT,ch}^{\mathrm{max,mcr}} \times t \quad (6)$$

$$P_{BT,ch}^{\mathrm{max,mcc}} = \frac{N_{BT} I_{max} V_{nom}}{1000} E_{BT,ch}^{\mathrm{max,mcc}} = P_{BT,ch}^{\mathrm{max,mcc}} \times t \quad (7)$$

The actual maximum charge power is the minimum of the three limits divided by round-trip efficiency:

$$P_{BT,ch}^{\max} = \frac{Min\left(P_{BT,ch}^{\max,\mathrm{KiBaM}} P_{BT,ch}^{\max,\mathrm{mcr}}, P_{BT,ch}^{\max,\mathrm{mcc}}\right)}{\eta_{BT}} \; E_{BT,ch}^{\max} = P_{BT,ch}^{\max} \times t \tag{8}$$

The maximum discharge power is calculated from [18]:

$$P_{BT,dch}^{\max,\mathrm{KiBaM}} = P_{BT,dch}^{\max} = \frac{kQ_1 e^{-t} + Qkc(1 - e^{-kt})}{1 - e^{-kt} + c(k\Delta t - 1 + e^{-kt})} \; E_{BT,dch}^{\max} = P_{BT,dch}^{\max} \times t \tag{9}$$

The battery wear cost (Cost_BT_Wear [$/kWh]) represents the marginal cost of each kWh cycled through the battery [24][25][26]:

$$\mathrm{Cost}_{BT}^{Wear} = \frac{R_{BT} \times N_{BT}^{total}}{N_{bat} \times Q_{lifetime} \times \sqrt{\eta_{BT}}} \tag{10}$$

Where R_BT is the replacement cost per kWh, N_BT_total is the total battery storage capacity, N_bat is the number of battery units, and Q_lifetime is the lifetime throughput in kWh.
Battery energy for the next time-step is updated as [A1]:

$$E_{BT}(t+1) = (1 - \delta) \times E_{BT}(t) + \eta_{BT} \times E_{BT}^{ch}(t) - \frac{E_{BT}^{dch}(t)}{\eta_{BT}} \tag{37}$$

Where delta is the hourly self-discharge rate.

17.4 Backup (Diesel or Gasoline) Generator Model

The Backup generator (BG) is modeled as a backup dispatchable source. The total hourly operating cost of the DG is calculated from its fuel consumption using a linear fuel curve [26][27][28]:

$$\mathrm{Cost}_{DG} = b \times P_{DG}^{Nominal} \times \mathrm{Cost}_{fuel} + \frac{R_{DG} \times P_{DG}^{Nominal}}{TL_{DG}} + MO_{DG} + a \times \mathrm{Cost}_{fuel} \tag{11}$$

Where a [L/hr/kW output] and b [L/hr/kW rated] are the slope and intercept of the fuel consumption curve (defaults a = 0.2730, b = 0.0330 from [11]), P_DG_Nominal [kW] is the total rated DG power, Cost_fuel [$/L] is the diesel fuel cost, R_DG [$/kW] is the replacement cost, TL_DG [hours] is the DG lifetime, and MO_DG [$/op.h] is the operations and maintenance cost.

17.5 Wind Turbine Model

Hub-height wind speed is extrapolated from anemometer height h0 to hub height h_hub using the power-law wind shear model:

$$\mathrm{V_hub} = \mathrm{V_anemometer} * (\mathrm{h_hub} / \mathrm{h0})\text{^}\mathrm{alpha_wind_turbine} \tag{WT}$$

Where alpha_wind_turbine is the friction coefficient (default 0.14 for open terrain). Wind turbine output follows a piecewise power curve: zero below cut-in speed (v_cut_in = 2.5 m/s), linear ramp from cut-in to rated speed (v_rated = 9.5 m/s), constant rated power from rated to cut-out speed (v_cut_out = 25 m/s), and zero above cut-out.

17.6 Energy Management Strategy (EMS)

SAMA implements an advanced load-following dispatch strategy [29][30] that operates over all 8,760 hours of the year. In each time-step t, when renewable generation (E_RE) exceeds load demand (E_load), excess energy is used to charge the battery and sell to the grid:

$$E_{BT}^{ch}(t) = \min\left(E_{BT}^{empty}, E_{RE}(t) - \frac{E_{load}(t)}{\eta_{inv}}\right) and \; E_{BT}^{ch}(t) = \min\left(E_{BT}^{ch}(t), E_{BT,ch}^{max}\right) \tag{12}$$

$$E_{BT}^{empty} = \frac{E_{\mathrm{BT}}^{\max} - E_{BT}}{\eta_{BT}} \quad (13)$$

$$E_{AC}^{sur} = \eta_{inv} \times (E_{RE}(t) - E_{BT}^{ch}(t)) - E_{load}(t) \quad (14)$$

$$E^{sell}(t) = min(E_{AC}^{sur}, E^{sell,max}) \; and \; E^{sell}(t) = \min\,(\max(0, E_{inv}^{max} - E_{load})), E^{sell}(t)) \quad (15)$$

When load exceeds renewable generation, SAMA selects the dispatch order among battery, DG, and grid based on marginal cost comparison. Six prioritization scenarios are defined based on the relative values of the grid buy price (C_buy), DG cost (Cost_DG), and battery wear cost (Cost_BT_Wear):
Scenario 1 (Grid, DG, Battery): C_buy <= Cost_DG and Cost_DG <= Cost_BT_Wear
Scenario 2 (Grid, Battery, DG): C_buy <= Cost_BT_Wear and Cost_BT_Wear < Cost_DG
Scenario 3 (DG, Grid, Battery): Cost_DG < C_buy and C_buy <= Cost_BT_Wear
Scenario 4 (DG, Battery, Grid): Cost_DG < Cost_BT_Wear and Cost_BT_Wear < C_buy
Scenario 5 (Battery, DG, Grid): Cost_BT_Wear < Cost_DG and Cost_DG < C_buy
Scenario 6 (Battery, Grid, DG): All remaining cases
The loss of power supply probability (LPSP) constraint tracks unmet energy Ens(t) across the year:

$$LPSP = \frac{\sum_t^{8760} Ens(t)}{\sum_t^{8760} E_{load}(t)} \quad (53)$$

The renewable fraction (RF) constraint ensures the minimum renewable share is met [6]:

$$RF = 1 - \frac{\sum_t^{8760} E_{non\,renewable}(t)}{\sum_t^{8760}(E_{load}(t) - ENS(t))} \quad (54)$$

17.7 Economic Model

The real discount rate i is derived from the nominal discount rate i' and expected inflation rate f [A1]:

$$i = \frac{i' - f}{1 + f} \quad (38)$$

The Net Present Cost (NPC) is the total lifecycle cost of the system discounted to present value [31]:

$$NPC = C_I + \frac{C_R + C_{MO} + C_F - C_S + C_G}{(1+i)^n} \quad (39)$$

The initial investment cost C_I includes all components [5]:

$$C_I = C_{PV} \times P_{PV}^{total} + C_{DG} \times P_{DG}^{total} + C_{BT} \times N_{total}^{BT} + C_{Inv} \times P_{Inv} + C_{CH} + C_{Eng} \quad (40)$$

Engineering costs C_Eng aggregate installation, overhead, sales and marketing, permitting and inspection, electrical and structural balance-of-system, supply chain, profit, and sales tax from NREL PV cost benchmarks [3]. Replacement, O&M, and fuel costs are:

$$C_R = C_{PV}^R \times P_{PV}^{total} + C_{DG}^R \times P_{DG}^{total} + C_{BT}^R \times N_{total}^{BT} + C_{Inv}^R \times P_{Inv} + C_{CH}^R \quad (41)$$

$$C_{MO} = C_{PV}^{MO} \times P_{PV}^{total} + C_{DG}^{MO} \times P_{DG}^{total} + C_{BT}^{MO} \times N_{total}^{BT} + C_{Inv}^{MO} \times P_{Inv} + C_{CH}^{MO} \quad (42)$$

$$C_F = \sum_0^{8760} Fuel\ cost\ per\ liter \times q \quad (43)$$

Salvage values for each component are calculated based on remaining lifetime [32]:

$$C_S = S_{PV} + S_{DG} + S_{BT} + S_{inv} + S_{CH} \text{ in which } S_{comp} = R_{comp} \times \frac{n_{rem}}{n_{comp}} \quad (44)$$

Total grid cost over the project lifetime includes fixed annual charges, monthly service charges, and variable hourly buy/sell costs [A1]:

$$C_G = C_{Grid}^{Fixed,Annual} + \sum_{0}^{12} C_{Grid}^{Fixed,monthly} + \sum_{0}^{8760} C_{buy} \times P_{buy} - \sum_{0}^{8760} C_{sell} \times P_{sell} \quad (45)$$

The Capital Recovery Factor (CRF) annualizes the NPC for LCOE calculation [33]:

$$CRF(i, N) = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (46)$$

The Levelized Cost of Energy (LCOE) is then [33]:

$$LCOE = \frac{CRF \times NPC}{\sum_0^{8760}(Eload - Ens + P_{sell})} \quad (47)$$

When multi-objective optimization is enabled (EM = 1), SAMA also minimizes Levelized Emission (LE), defined similarly to [34]:

$$LE = \frac{\sum_0^{8760} Non\ Grid\ Emissions\ (t) + \sum_0^{8760} Grid_{Emissions}(t)}{\sum_0^{8760}(E_{load}\ (t) - Ens(t))} \quad (49)$$

DG fuel consumption and emissions are calculated from the linear fuel curve [26]:

$$q = a \times P_{DG} + b \times P_{DG}^{Nominal} \quad (51)$$

Grid emissions are proportional to electricity imported:

$$Grid\ Emissions\ = E_{buy} \times EF_{Grid} \quad (52)$$

The multi-objective function combines NPC and LE, with penalties applied when constraints are violated [A1]:

$$Z = NPC + EM \times LE + penalties \quad (55)$$

Table 2 summarizes the penalty conditions applied to infeasible candidate solutions:

Penalty Condition	Reason
DC to AC ratio > 1.99 * (P_Inv + P_DG + P_buy_max)	DC to AC ratio must not exceed 2 for a feasible inverter design
LPSP > LPSP_max	System does not supply sufficient power to meet reliability constraint
RF < RE_min	Renewable fraction is below the user-specified minimum
I_Cost > Budget	Initial capital cost exceeds the user-specified budget limit

Table 2. Penalty conditions in SAMA optimization (from [A1]).

17.8 Validation Against HOMER Pro

SAMA results have been cross-validated against HOMER Pro across two geographically distinct U.S. locations: Sacramento, California and New Bern, North Carolina [A1]. Validation used identical input data (same load profiles from OpenEI [2], meteorological data from NSRDB [1], and equivalent pricing). The PSO optimizer was run with MaxIt = 101 and nPop = 100, repeated 10 times to assess convergence. Convergence was achieved by iteration 70 on the first run and as early as iteration 20 on subsequent runs, demonstrating consistent optimization behavior. Final system configurations, LCOE, NPC, energy distribution, and battery state-of-charge profiles from SAMA showed close agreement with HOMER Pro across all tested system types (PV+Grid, PV+BT+Grid, PV+BT+DG+Grid, PV+BT+DG, PV+BT).

Minor differences between SAMA and HOMER Pro outputs are attributable to differences in dispatch strategy implementation, pricing methodology rounding, and the fact that SAMA's load-following EMS was developed independently with a stronger emphasis on minimizing power shortages. A key advantage of SAMA is its novel top-down pricing methodology [A1], which is not available in HOMER Pro and allows sensitivity analysis on total PV system cost per kW without requiring itemized component cost inputs.

For the off-grid PV+DG+Battery configuration in Sacramento, the optimizer found that maintaining a grid connection (LCOE = $0.0838/kWh) is significantly more economical than full grid defection ($0.3674/kWh). Similar conclusions apply to New Bern. This capability makes SAMA particularly useful for grid-defection analysis [A2].

18. References

The following references are cited in this documentation and in the SAMA technical framework described in Section 17. For access to SAMA-related publications, contact the FAST Research Group at Western University (see Section 19).

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19. SAMA Publications and Further Reading

The following peer-reviewed publications describe SAMA's development, validation, and application across a range of techno-economic energy studies. Users who wish to cite SAMA in academic work should reference these papers. If you need copies of any of these publications, please contact the developers at the FAST Research Group, Western University.

[A1] Sadat, S.A., Takahashi, J. and Pearce, J.M., 2023. A Free and open-source microgrid optimization tool: SAMA the solar alone Multi-Objective Advisor. Energy Conversion and Management, 298, p.117686. https://doi.org/10.1016/j.enconman.2023.117686

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For the most up-to-date list of SAMA publications and to request preprints or reprints, visit the FAST Research Group GitHub at https://github.com/Sas1997/SAMA or contact the corresponding author directly.

Appendix A: Where to Find Input Data for SAMA

This appendix provides practical guidance on where to obtain real-world data for each major input category in SAMA. For each parameter group, the purpose, typical values, and recommended data sources are described so that you can replace the bundled defaults with values appropriate for your specific project location and system.

A.1 Financial and Economic Parameters

Nominal Discount Rate: Used to discount future cash flows to present value in NPC calculations. Check the Federal Reserve primary credit rate at federalreserve.gov or financial data sites such as YCharts for U.S. projects. For non-U.S. projects, use your local central bank rate. Typical values are 2 to 5%. Input as a percentage (e.g., 4 for 4%).

Expected Inflation Rate: Accounts for anticipated price rises over the project lifetime. For U.S. projects, refer to the Bureau of Labor Statistics CPI reports at bls.gov, or check Federal Reserve projections at federalreserve.gov. For other countries, use national statistics agencies. Input as a percentage (e.g., 3 for 3%).

RE Incentives / Federal Tax Credit: In the United States, the Investment Tax Credit (ITC) provides a 30% credit for solar under the Inflation Reduction Act. Check the Database of State Incentives for Renewables and Efficiency (DSIRE) at dsireusa.org for current federal and state programs. For non-U.S. projects, consult your national energy ministry or IRENA incentive databases.

A.2 System Constraints

LPSP (Loss of Power Supply Probability): Set to 0% for critical loads (medical, emergency). For non-critical off-grid applications, 1 to 5% is typical. IEEE and HOMER Energy publications provide benchmark values for different application types.

Minimum Renewable Energy Fraction: Align with local Renewable Portfolio Standards (RPS) or your project sustainability target. DSIRE at dsireusa.org lists U.S. state RPS requirements. IRENA reports provide international benchmarks.

Net Metering Cap (kW): Check your utility net metering policy at dsireusa.org. Common U.S. residential caps range from 10 to 100 kW. Non-U.S. users should check local grid interconnection regulations.

Rooftop Area Limit: Measure usable roof area using Google Earth, satellite imagery, or a professional site survey. Subtract areas for chimneys, vents, setbacks (typically 3 feet from roof edges), and shading. As of 2025, high-efficiency monocrystalline panels require approximately 7 to 9 m2 per kW installed including spacing.

A.3 Weather and Meteorological Data (METEO.csv)

The bundled METEO.csv is for London, Ontario. For any other location, download a new file. The file must contain GHI, DNI, DHI, Temperature, Wind Speed, and Pressure columns with two header rows followed by 8,760 hourly rows.

Primary source: National Solar Radiation Database (NSRDB) at nsrdb.nrel.gov. Provides free hourly weather data for the U.S., Canada, and many other countries. Select the SAM/CSV export format. This is the recommended source for SAMA.

Alternative: NASA POWER at power.larc.nasa.gov provides global coverage for any coordinates. Export to CSV and reformat columns to match the METEO.csv structure.

POA irradiance (for G_type = 2): Use PVWatts at pvwatts.nrel.gov or SAM at sam.nrel.gov to compute hourly plane-of-array irradiance from your tilt and azimuth. Save as Irradiance.csv (one column, 8,760 rows, no header, W/m2).

Tip: When downloading from NSRDB, select the Typical Meteorological Year (TMY) dataset rather than a single calendar year. TMY data is specifically designed for long-term energy system analysis.

A.4 Electrical Load Data (Eload.csv)

Eload.csv requires one column of 8,760 hourly load values in kW with no header. The bundled file is for a London, Ontario residential building. Options to replace it:
Smart meter data: Most utilities provide 15-minute or hourly interval data through their customer portal (Green Button data in North America). Download and aggregate to hourly kW values.
From bills: Use SAMA load modes 3 to 8 (monthly or annual average inputs) to let SAMA generate a synthetic 8,760-hour profile from your monthly kWh totals.
Open databases: OpenEI at openei.org provides representative commercial and residential load profiles by building type and U.S. climate zone. DOE Commercial Reference Building models are also freely available.

A.5 Thermal Load Data (house_load.xlsx) for Heat Pump

house_load.xlsx must have hourly heating load (Hload) in column 2 and cooling load (Cload) in column 3, 8,760 rows each, in kW. Options:
EnergyPlus: Free DOE building simulation software at energyplus.net. Build a model with local weather and export hourly heating/cooling loads. Most accurate approach.
RETScreen: Free software from Natural Resources Canada at nrcan.gc.ca. Provides monthly heating and cooling demand that can be distributed hourly.
Manual entry: In sama-config, set Tload_type to 2 or higher and enter monthly average loads. SAMA generates a synthetic hourly profile.

A.6 Solar PV Parameters

Module efficiency and temperature coefficient: From the manufacturer datasheet. Use EnergySage (energysage.com) or the California Energy Commission (CEC) equipment list to compare modules. Leading 2025 monocrystalline modules range from 20 to 25% efficiency.
Capital and O&M costs: NREL cost benchmark reports at nrel.gov, SEIA quarterly market reports at seia.org, or installer quotes via EnergySage. NREL reports residential PV system costs of USD 2,500 to 3,500 per kW installed and O&M of USD 20 to 40 per kW per year for 2025.
RE incentives: Federal ITC (30%) and state programs at dsireusa.org. Factor incentives into capital cost or use the RE_incentives_rate field.
Azimuth and tilt: For the Northern Hemisphere, south-facing (180 degrees) at a tilt equal to your site latitude maximizes annual yield. Use PVWatts (pvwatts.nrel.gov) to optimize for your specific location.

A.7 Wind Turbine Parameters

Wind speed data: Included in the METEO.csv from NSRDB. NASA POWER (power.larc.nasa.gov) also provides global wind data at multiple heights.
Turbine specs (hub height, cut-in/rated/cut-out speeds): From the manufacturer datasheet. Bergey Windpower, Northern Power Systems, and other small turbine manufacturers publish detailed performance curves. Typical small turbine values: cut-in 2.5 to 3.5 m/s, rated speed 9 to 13 m/s, cut-out 25 m/s.
Friction coefficient (wind shear exponent): Use 0.14 for open flat terrain, 0.20 for suburban, 0.30 for forested or urban. WindExchange at windexchange.energy.gov provides terrain roughness guidance.
Capital costs: NREL Distributed Wind Market Report at nrel.gov/wind. Small residential turbines (under 100 kW) cost approximately USD 4,000 to 8,000 per kW installed in 2025.

A.8 Battery Storage Parameters

Capital cost: NREL Annual Technology Baseline at nrel.gov, SEIA reports, EnergySage battery quotes. Li-ion installed costs average USD 200 to 400 per kWh in the U.S. for 2025. Lead-acid is lower cost but requires more frequent replacement.

Round-trip efficiency and SOC limits: From manufacturer datasheet. Li-ion: 90 to 95% round-trip efficiency. Lead-acid: 70 to 85%. Battery University at batteryuniversity.com provides practical guidance. Most manufacturers recommend keeping lead-acid above 50% SOC.

Lifetime throughput: From manufacturer cycle life tables. NREL and PNNL publish battery degradation models for Li-ion. Combine rated cycle count with depth of discharge per cycle to estimate total kWh throughput.

A.9 Diesel Generator Parameters

Fuel cost per liter: U.S. EIA weekly petroleum reports at eia.gov/petroleum, GasBuddy at gasbuddy.com, or GlobalPetrolPrices at globalpetrolprices.com. Convert from dollars per gallon to dollars per liter by dividing by 3.785.

Fuel curve coefficients (a and b): From the manufacturer fuel consumption data sheet. HOMER Energy documentation and Woodstock Power/Generator Source websites publish consumption curves for common generator sizes. Typical: a = 0.24 to 0.44 L/hr per kW output, b = 0.01 to 0.11 L/hr per kW rated.

CO2 emissions: U.S. EIA default is 2.68 kg CO2 per liter of diesel. The Climate Registry at theclimateregistry.org and EPA emission factor documents provide updated values.

A.10 Heat Pump Parameters

SAMA includes built-in performance models for Bosch and Goodman brand heat pumps. Select the brand that most closely matches your planned installation. If using a different brand, choose the model with the most similar efficiency ratings.

Rated capacity: Determined by Manual J heating/cooling load calculations. HVAC contractors perform Manual J per ACCA standards. The required capacity is typically 15 to 30 BTU/hr per square foot of floor area depending on climate.

Capital cost: EnergySage, Angi, and Home Depot provide current quotes. For 2025, mini-split systems cost approximately USD 3,000 to 8,000 installed for a 12,000 BTU unit. Normalize to dollars per 1,000 BTU/hr for SAMA input. Federal IRA rebates (up to USD 2,000) and state incentives at dsireusa.org can reduce effective cost.

A.11 Electric Vehicle Parameters

Battery capacity, range, and charging specs: EV-Database.org, InsideEVs (insideevs.com), or the U.S. EPA at fueleconomy.gov. Manufacturer specification sheets provide usable battery capacity, AC charging power, and efficiency ratings.

Battery replacement cost: Recurrent Auto (recurrentauto.com) tracks real-world EV battery health and replacement cost estimates. Manufacturer warranty documentation covers replacement terms. As of 2025, battery costs are approximately USD 100 to 150 per kWh.

Daily trip distance: From your own odometer records or U.S. DOT National Household Travel Survey (NHTS) at nhts.ornl.gov. U.S. average daily vehicle travel is approximately 50 to 65 km.

Charging schedule: Based on your own commuting pattern. For default values, use the U.S. DOT NHTS or INRIX traffic data for regional averages.

A.12 Grid and Utility Rate Parameters

Grid rate parameters have the largest impact on SAMA economic results. Always use your actual utility tariff rather than national averages.

Electricity rate schedule: Download from your utility website or OpenEI Utility Rate Database at openei.org/apps/USURDB. Covers most U.S. and many Canadian utilities. Identify your plan type (flat, tiered, TOU, ULO) and enter the matching structure in SAMA.

Current electricity prices: From your most recent utility bill. U.S. EIA publishes monthly state average prices at eia.gov/electricity. Canadian prices from the Canadian Energy Regulator at cer-rec.gc.ca.

Electricity price escalation: Review historical rate filings on your state Public Utility Commission (PUC) website. EIA Annual Energy Outlook provides regional long-term projections. U.S. average is 2 to 3% per year historically.

Net metering policy: DSIRE at dsireusa.org for state rules. Your utility interconnection agreement specifies the export compensation rate. Some states have moved to reduced net metering rates (e.g., California NEM 3.0).

Monthly fixed charge and taxes: From your utility bill line items. Monthly service charge is listed as a fixed dollar amount. Sales tax and per-kWh surcharges are itemized separately.

Natural gas rates: From your gas utility bill or U.S. EIA natural gas reports at eia.gov/naturalgas. Canadian rates from provincial utility regulators. Identify the price per m3 or per therm and rate structure.

Tip: For TOU or ULO rate structures, contact your utility customer service or visit their website for the exact on-peak, mid-peak, and off-peak hour windows. These vary significantly by utility and season.

A.13 Summary of Recommended Data Sources

Input Category	Recommended Data Source
Weather (METEO.csv)	NSRDB (nsrdb.nrel.gov), NASA POWER (power.larc.nasa.gov)
POA Irradiance (Irradiance.csv)	PVWatts (pvwatts.nrel.gov), SAM (sam.nrel.gov)
Electrical load (Eload.csv)	Utility smart meter / Green Button data, OpenEI load profiles
Thermal load (house_load.xlsx)	EnergyPlus (energyplus.net), RETScreen (nrcan.gc.ca)
PV module specs and costs	Manufacturer datasheet, EnergySage, NREL cost benchmarks
PV incentives	DSIRE (dsireusa.org), IRS ITC documentation
Wind turbine specs and costs	Manufacturer datasheet, NREL Distributed Wind Market Report
Battery costs	NREL Annual Technology Baseline, SEIA reports, EnergySage
Battery specs	Manufacturer datasheet, Battery University (batteryuniversity.com)
Diesel fuel cost	EIA weekly petroleum (eia.gov), GasBuddy, GlobalPetrolPrices
Diesel emissions factors	U.S. EPA emission factors, The Climate Registry (theclimateregistry.org)
Heat pump costs	EnergySage, Angi, DSIRE for IRA rebates
EV specs and battery cost	EV-Database.org, InsideEVs, fueleconomy.gov, Recurrent Auto
Electricity rates	Your utility bill, OpenEI USURDB (openei.org), EIA
Net metering policy	DSIRE (dsireusa.org), utility interconnection agreement
Discount rate	Federal Reserve (federalreserve.gov), local central bank
Inflation rate	BLS CPI (bls.gov), Federal Reserve projections
Natural gas rates	Gas utility bill, EIA natural gas (eia.gov/naturalgas)